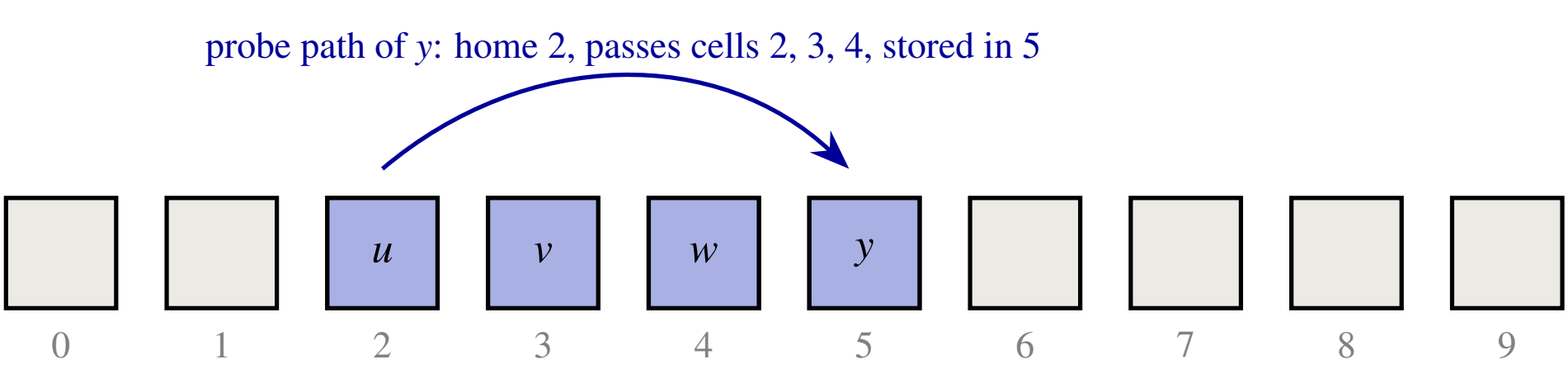
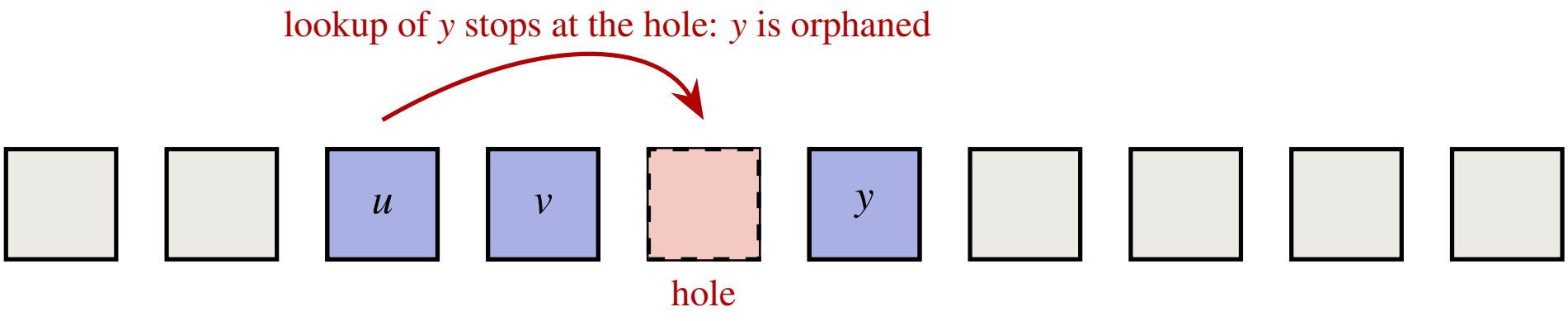


Dependency: $y \lhd x$ if the cell of x lies strictly before the cell of y on the probe path of y.
Here $y \lhd u$, $y \lhd v$, $y \lhd w$. A deletion order is safe iff every key is cleared before the keys it depends on (linear extension of $\lhd$).

Clearing w before y (not stack order):



Stack order is always safe: the newest key has no dependents, because a key's path crosses only cells that were filled before it. It is also the only order safe for every hash assignment (Thm 4b). The number of keys orphaned by clearing one key is a queue length minus one (Prop 5).